

A Dual-Layer Architecture for Asset-Centric RWA Tokenization on Blockchain

Umut Pekel

*Institute of Applied Mathematics
Middle East Technical University
Çankaya, 06800, Ankara, Turkey
umut.pekel@metu.edu.tr
0000-0002-7913-9616*

Oğuz Yayla

*Institute of Applied Mathematics
Middle East Technical University
Çankaya, 06800, Ankara, Türkiye
oguz@metu.edu.tr
0000-0001-8945-2780*

Abstract—Infrastructure assets often generate stable, policy-backed cash flows, yet existing investment instruments bundle asset exposure with corporate balance-sheet and governance risks. Existing RWA tokenization does not change this; it still treats an asset as a contract-based claim rather than as the asset itself. We argue that reducing property rights (rights tied to the asset) to rights tied to a contract is a fundamental limitation, and one that a smart contract alone cannot overcome. What is needed is an object-centric model in which each asset unit carries its own identity and rights natively. We propose a dual-layer architecture that separates legal asset anchoring from investor-facing cash-flow participation. The first layer anchors the asset identity and legal linkage. The second layer issues participation units with a mutable yield-beneficiary pointer, allowing custody and cash-flow entitlement to be held independently. The architecture targets policy-backed infrastructure assets, such as renewable energy under feed-in tariffs, where generation, validation, and settlement are already institutionally established. A token-bound dividend routing algorithm distributes revenues at the individual token level via parallel batched transactions. Cost projections show that micro-distributions across one million tokens can be executed at negligible marginal cost under parallel execution, compared to prohibitive gas expenses in sequential EVM-based models. The contribution is a blockchain-native design space and representation model that gives investors exposure to asset-level yields, not to the risks of a corporate wrapper.

Index Terms—Blockchain, infrastructure finance, asset-centric tokenization, renewable energy finance, property-right representation, policy-backed cash flows, direct asset participation

I. INTRODUCTION

Most real-world asset (RWA) tokenization efforts treat the blockchain as a new infrastructure layer onto which existing financial structures are directly migrated. We argue that this is insufficient: the main challenge is representation, not migration. When assets are brought on-chain, they should be redesigned to match blockchain's execution and programmability.

Current approaches preserve corporate wrappers, account-based balances, and aggregated ownership, creating a mismatch with native blockchain execution. This is particularly visible in infrastructure finance, where policy-backed cash flows are mediated through corporate entities that bundle unrelated risks.

The central idea is to shift from corporate exposure to asset-level participation via a dual-layer architecture: legal asset anchoring and investor-facing yield rights. This framework maps institutionally validated cash flows to programmable entitlements without introducing a new financial system.

a) Problem and Thesis: Infrastructure assets generate stable policy-backed cash flows, yet investment instruments bundle unrelated risks, raise participation thresholds, and rely on layered intermediation. Our thesis: blockchain enables direct asset participation, redefining ownership and cash-flow rights at the asset level.

b) Contributions: 1) Token-centric settlement treating each asset unit as an independent programmable entity. 2) Programmable yield routing decoupling custody from cash-flow rights via mutable beneficiary pointers. 3) System architecture for scalable, policy-backed revenue distribution using high-throughput ledgers.

II. RELATED WORK

Early work on blockchain in energy systems focused on peer-to-peer trading and certification mechanisms. More recent research has moved toward tokenization of real-world infrastructure assets, showing potential improvements in liquidity and financing efficiency [1]–[3]. However, most approaches replicate existing financial structures, embedding asset exposure within corporate or contractual layers.

The question of inclusivity has also been widely studied. Previous work shows that tokenization can lower entry barriers for small investors, but does not

resolve underlying asset risk or valuation challenges [5], [6], [8]. This suggests that accessibility alone is insufficient without a change in the way investment exposure is structured.

From a legal perspective, several studies argue that tokenized claims often provide only limited or indirect ownership unless supported by strong legal anchoring [9], [10]. Without such anchoring, tokens risk representing economic expectations rather than enforceable rights.

On the technical side, previous work highlights the importance of token standards and execution models. In particular, ERC-1155 has been shown to support more efficient batch processing and high-frequency settlement compared to earlier standards [11], [12]. However, most existing implementations remain account-based, aggregating balances at the user level rather than representing asset units as independent objects. However, ERC-1155 remains account-based: it aggregates balances per user and does not support per-token programmable beneficiary pointers. The proposed asset-centric model treats each token as an independent object with mutable cash-flow routing, enabling yield separation and collateralization at the unit level—capabilities not expressible in ERC-1155 without additional layering.

Architecturally, the closest work is the two-tier model proposed in [13], which introduces asset decomposition combined with enterprise-level aggregation. Although this improves composability, the aggregation step reintroduces exposure to corporate-level risks. In contrast, this paper adopts an asset-centric approach that avoids enterprise-level aggregation and focuses on directly representing asset-level policy-backed revenue streams.

III. CONCEPTUAL FOUNDATION

Traditional infrastructure finance bundles asset risk with corporate balance-sheet and governance risk. Our framework treats the asset and its cash-flow logic as the primary object. Instead of account-based aggregation, we adopt object-level representation where each unit carries rights and cash-flow logic. The blockchain serves as a deterministic execution layer, not a replacement for settlement systems.

IV. ASSET-CENTRIC REPRESENTATION OF PROPERTY AND YIELD RIGHTS

Most tokenization models follow an account-based design inherited from traditional finance. In these systems, assets are represented as balances associated with accounts, and asset identity is treated as a secondary attribute.

The proposed framework takes a different approach. It treats the asset itself, rather than the account, as the primary unit of representation. A unit of infrastructure—such as installed capacity, leasable space, or any revenue-generating component—is modeled as a distinct object with associated rights. These rights may include ownership reference, yield entitlement, and cash-flow routing logic.

In this model, fungibility is not the starting point. Instead, each unit is initially defined as a specific object, and fungibility is introduced only at the investor-facing layer where appropriate. This allows the system to preserve asset-level detail while still supporting scalable participation.

In the proposed architecture, this leads to a clear separation of roles. The asset anchor layer preserves identity, legal linkage, and institutional reference. It is not designed for circulation. The investor-facing layer, on the other hand, consists of transferable participation units that represent claims on the asset's revenue stream. What remains fixed is the definition of the asset; what moves is the economic participation right.

Existing token standards fall into two categories: fungible tokens (loss of identity) and NFTs (no native yield rights). Our model combines identity, legal reference, and programmable financial rights in one object.

The proposed framework introduces a different class of representation. Each asset unit is modeled as an object that simultaneously carries identity, legal reference, and programmable financial rights.

Although such structures exist naturally in real-world assets, they are not adequately represented in current blockchain-based tokenization systems. The main contribution of this work is to define an abstraction that captures this structure in a way that is consistent not only with its underlying economic and legal characteristics, but also with the execution and programmability model of blockchain systems.

The value of blockchain in this context lies not in data storage, but in programmable and deterministic execution.

V. PROPOSED ARCHITECTURE

The proposed architecture is based on a strict separation between legal asset anchoring and investor-facing cash-flow participation. This separation reflects the economic thesis of the paper: infrastructure investment should be organized around project-level revenue rights rather than around the total risk envelope of a corporate issuer. To implement this principle, the architecture contains two conceptually distinct layers.

The first layer is the *asset anchoring layer*. Here, the relevant infrastructure project is isolated through an appropriate legal wrapper, such as a project-specific entity

or comparable ring-fenced structure, and represented on-chain through a non-transferable anchor object. Its role is to preserve the identity of the productive asset, to bind relevant legal and technical metadata, and to define the institutional reference point from which investor-facing claims derive legitimacy.

The second layer is the *cash flow participation layer*. In this layer, fungible investor-facing units are issued to represent economically meaningful shares in the distributable revenue stream generated by the anchored asset. These units do not denote generalized corporate ownership. Instead, they express a rule-based entitlement to a portion of project-level cash flows under the conditions defined by the relevant contractual and regulatory framework. Their purpose is to enable broad participation, transferability, and scalable distribution without destroying the asset specificity established at the anchor layer.

In such environments, the state or authorized market operator already defines how output is measured, cleared, settled, and paid. The blockchain layer does not replace this institutional order. Rather, it receives institutionally validated settlement events and translates them into investor-level distribution logic. The technology therefore functions as an execution layer for legally and regulatorily grounded entitlements, not as an autonomous source of economic truth.

The architecture also allows the separation of custody from benefit. Because investor-facing yield rights are represented in programmable form, the destination of cash-flow benefits may be specified independently of the mere possession of the participation unit. This design opens the way for more refined financial structures, including redirected income rights, collateralized revenue streams, and differentiated beneficiary arrangements, without requiring full alienation of the underlying participation claim.

A reference implementation can be constructed using a hybrid architecture: (i) a permissioned ledger (e.g., Hyperledger Fabric) for asset anchoring and regulatory control, and (ii) a high-throughput public or semi-permissioned execution layer (e.g., Solana, Aptos) for scalable micro-settlement.

Communication between layers occurs via a lightweight oracle bridge that relays settlement events—already validated by the institutional market operator—from the permissioned registry to the execution layer. The execution layer maintains no independent notion of asset identity; it only processes yield distribution instructions against pre-issued Y-BITs.

As summarized in Table I, while current standards such as ERC-3643 provide robust compliance through identity registries, they remain bound to account-based

TABLE I
TECHNICAL COMPARISON OF TOKENIZATION MODELS

Feature	ERC-1155	ERC-3643	UTXO	Y-BIT (Ours)
Representation	Account-based	Account-based	Input/Output	Asset-centric Object
Per-token yield beneficiary	No	No	Manual	Yes (native)
Ownership vs. Cash-flow decoupling	No	No	Via scripts	Yes (mutable pointer)
Batch settlement	Yes	Limited	No	Yes (parallel)
Risk isolation	Bundled	Issuer-level	Transactional	Asset-level

state updates, which limits the distribution of granular yields. In contrast, our proposed Y-BIT architecture draws inspiration from the parallelizability of UTXO models but evolves it into an object-based representation. This transition from “who owns what” to “what rights the asset carries” allows for native yield routing and strict risk isolation, addressing the fundamental requirements for large-scale infrastructure tokenization.

a) State Authority Across Layers: The permissioned asset anchor layer is the authoritative source of truth for asset identity, legal linkage, and issuance constraints. The execution layer is the authoritative source of truth for Y-BIT ownership and the `yieldBeneficiary` field used in settlement. The oracle bridge does not create a new asset state; it only relays institutionally validated settlement events and issuance-permitted references from the anchoring layer to the execution layer. In this way, legal identity remains anchored in the permissioned registry, while economic participation and yield routing remain authoritative on the execution layer.

b) Execution-Layer Token State: Each Y-BIT instance maintains two minimal fields:

- `owner`: the custody holder of the token,
- `yieldBeneficiary`: the programmable destination for dividend distribution.

The `yieldBeneficiary` field is mutable via `RedirectYield` and determines where settlement payments are sent, decoupling custody from cash-flow entitlement.

All Y-BIT tokens issued against a given anchored asset, denoted by the set $\mathcal{Y}$, are maintained on the execution layer together with their `owner` and `yieldBeneficiary` fields. Consequently, the execution layer is the authoritative source of truth for investor-facing token state, while the permissioned anchoring layer remains authoritative for asset identity and legal linkage.

c) Implementation Note: The algorithm expresses the logical distribution at the level of individual token instances. In a deployment on a parallel-execution ledger (e.g. Solana, Aptos, or a custom app-chain), the set of tokens is partitioned into batches. Each batch is processed in a separate transaction, and these transactions are submitted concurrently. The ledger’s

Algorithm 1 Token-Bound Dividend Routing Logic (Y-BIT Protocol)

Require: Oracle $\mathcal{O}$, Compliance Registry $\mathcal{K}$,
Require: Y-BIT Set $\mathcal{Y}$, Capacity S

- 1: $Tariff_{FIT} \leftarrow$ Policy-Fixed Rate
- ▷ Oracle submits Generation (e_t) and Period Costs ($opex_t$)
- 2: **function** EXECUTESSETTLEMENT($o, e_t, opex_t$)
- 3: **if** $o \notin \mathcal{O}$ **then return reject**
- 4: **end if**
- 5: Verify signature of data bundle ($e_t, opex_t$)
- 6: $GrossRevenue \leftarrow e_t \times Tariff_{FIT}$
- 7: $NetRevenue \leftarrow GrossRevenue - opex_t$
- 8: **if** $NetRevenue \leq 0$ **then return skip**
- 9: **end if**
- 10: $YieldPerWatt \leftarrow NetRevenue / S$
- ▷ Distribution logic expressed per token; actual execution uses batched parallel transactions
- 11: **for each** $ybit_i$ **in** $\mathcal{Y}$ **do**
- 12: $Beneficiary \leftarrow ybit_i.yieldBeneficiary$
- 13: DISTRIBUTE($Beneficiary, YieldPerWatt$)
- 14: **end for**
- 15: **end function**
- 16: **function** REDIRECTYIELD($tokenID, newAddress$)
- 17: **if** $msg.sender \neq \mathcal{Y}.ownerOf(tokenID)$ **then**
- 18: **return reject** (Unauthorized)
- 19: **end if**
- ▷ Enforce KYC on the new destination
- 20: **if not** $\mathcal{K}.isVerified(newAddress)$ **then return reject** (Compliance Fail)
- 21: **end if**
- 22: $\mathcal{Y}[tokenID].yieldBeneficiary \leftarrow newAddress$
- 23: Emit YieldRedirected($tokenID, newAddress$)
- 24: **end function**

parallel execution engine processes them simultaneously, achieving an effective cost of $O(S / batchSize)$ and amortizing overhead across the entire supply. For a facility with 10^6 tokens and a batch size of 10^4 , this yields 100 parallel transactions, each costing ≈ 0.0001 USD, making micro-distribution economically viable.

d) Parallel Execution and State Access: The proposed Y-BIT protocol is designed to leverage the parallel execution capabilities of high-performance ledgers (e.g., Solana’s Sealevel or Aptos’s Block-STM). Unlike traditional Ethereum Virtual Machine (EVM) models that suffer from sequential state contention, our asset-centric objects allow for non-overlapping state access.

By defining each atomic asset unit as a standalone program-owned account, the runtime can identify independent transactions that do not share the same writable accounts. This enables the `ExecuteSettlement` function (see Algorithm 1) to run concurrently across multiple compute units, theoretically increasing the throughput linearly with the number of hardware cores. This implementation-specific advantage transforms the “Design Space” into a high-throughput settlement en-

gine suitable for massive-scale RWA deployments.

VI. IMPLEMENTATION STRATEGY AND MARKET INTEGRATION

a) Architectural Requirements for High-Frequency Settlement: The economic viability of high-frequency micro-distributions imposes strict architectural constraints. Legacy execution environments, such as Ethereum Mainnet, suffer from prohibitive transaction overhead, rendering micro-dividend settlement economically infeasible. Viable deployment therefore requires an execution layer with negligible marginal cost and native transfer interception, so that compliance predicates can be enforced atomically at the protocol level prior to settlement. This architectural requirement eliminates the reliance on external compliance wrappers and enables a scalable, regulation-aware distribution of asset-backed cash flows.

TABLE II
SETTLEMENT COST COMPARISON (1M TOKENS)

Architecture	Cost	Feasibility
EVM sequential loop	$\sim \$50,000$	Infeasible
EVM + batching (single tx)	$\sim \$5,000$	Infeasible
Parallel L1 (batched, sequential txs)	$\sim \$100$	Marginal
Parallel L1 (batched, concurrent txs)	$\sim \$0.01\text{--}\1	Feasible

Cost estimates assume ≈ 0.0001 USD per transaction for parallel L1 (e.g. Solana mainnet fees).

b) Market Integration: The Pool-Based Advantage: Integration with physical electricity markets is designed to minimize financial risk. As illustrated in Figs. 1 and 2, conventional frameworks typically rely on bilateral agreements or vertically coordinated chains. In these models, revenue certainty is heavily dependent on counterparty creditworthiness and spot market volatility, creating a risk profile unsuitable for stable bond-like tokenization.

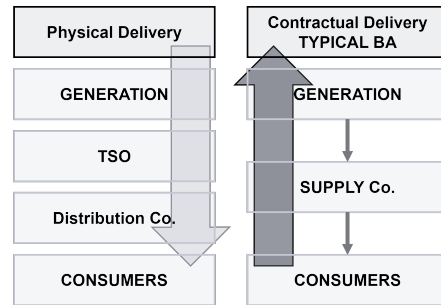


Fig. 1. Prevalent Market Framework: Bilateral and vertically coordinated settlement involving counterparty risk.

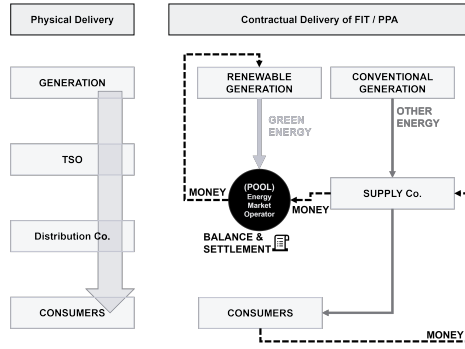


Fig. 2. Intermediate Framework: Bilateral agreements within a FIT context.

In contrast, our proposed framework (Fig. 3) integrates renewable producers into a centralized, **Pool-Based Financial Clearing Mechanism**. A distinguishing feature of this model is that FIT costs are typically socialized across the entire electricity consumer base. Funds collected from end-users are aggregated into a central clearing pool managed by a settlement authority, transforming the revenue model from a fragile B2B contract into a broad-based, consumer-funded ecosystem. From the perspective of token holders, this design significantly reduces the risk of a single-point counterpart. Rather than relying on the balance sheet of an individual utility, revenues are backed by the aggregate electricity consumption of the regulated market.

In such systems, revenue generation, validation, and clearing are institutionally defined and operational. The contribution of this work lies in how these already-cleared cash flows are represented and executed at the investor level, through a programmable and asset-specific settlement mechanism.

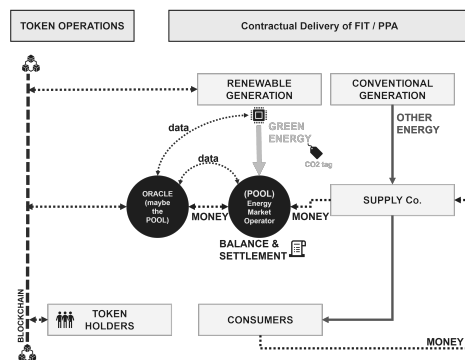


Fig. 3. Proposed Framework: Centralized, pool-based settlement decoupling revenue from bilateral risk.

Importantly, the proposed architecture does not introduce a new revenue model, but overlays an existing

class of policy-backed settlement systems. In several jurisdictions, renewable energy markets already operate under centralized clearing mechanisms in which generation is metered, validated by a market operator, and settled according to predefined tariff schemes.

The framework is built directly on such environments. Rather than redesigning the financial logic of these systems, it introduces a tokenized representation and execution layer that maps already-cleared revenues to programmable investor entitlements.

c) Instrument Definition: The Yield-Bearing Infrastructure Token: To bridge the gap between static asset ownership and recurring revenue distribution, we introduce the *Yield-Bearing Infrastructure Token (Y-BIT)*. Rather than being a speculative crypto-asset, the Y-BIT functions as a digital abstraction of a "Revenue Participation Note." It operates by binding the SPV's revenue directly to IoT meter readings and autonomously calculating dividends, thereby bypassing the administrative friction associated with traditional corporate boards.

d) Market Rationale: Gross Settlement vs. Netting: Crucially, the reliability of the Y-BIT depends on the regulatory design. Zahedi [4] highlights that "Net Metering" schemes introduce behavioral risk, as revenue fluctuates with the consumer's consumption. To mitigate this, our framework targets "Gross Settlement" markets, where revenues are computed strictly from metered generation delivered to the grid. This ensures a clear mapping from verified physics to financial yield, as summarized in Table III.

TABLE III
COMPARISON OF ENERGY TOKENIZATION MODELS

Feature	Market-Based Model	Proposed FIT-Anchored Model
Revenue Source	Spot market pricing	Policy-backed tariff / CfD settlement
Investor Risk	High exposure to price/demand volatility	Risk dominated by regulatory/operational factors
Financial Analogy	Equity-like, variable returns	Yield claim on infrastructure cash flows

e) Handling Regulatory Variants: Finally, we acknowledge that support schemes differ in enrollment rules; some lock terms at commissioning, while others allow periodic transitions between support and merchant exposure. The proposed framework treats these choices as external scheme parameters rather than as hard-coded logic. The on-chain architecture does not attempt to replicate complex institutional clearing laws; instead, it strictly binds cleared settlement events to distribution rules via authorized oracles. This design choice ensures that the system remains robust even

if regulatory parameters (e.g., tariff rates or eligibility periods) are updated off-chain.

f) Indicative Feasibility Perspective: To provide an indicative feasibility perspective, consider a deployment scenario involving 10^6 atomic units (1 Watt tokens). Under a parallel-execution architecture (e.g. Solana, Aptos), settlement is structured as 100 concurrent transactions, each processing 10^4 tokens. These transactions are submitted together and can be included in the same block, leveraging the ledger's parallel execution engine. The total cost is ≈ 0.01 USD, making micro-distribution economically viable.

In contrast, an EVM-based sequential model would require iterating over each unit or account, leading to $O(10^6)$ gas cost ($\approx \$50,000$ at current prices), rendering micro-distributions economically infeasible.

This qualitative comparison highlights that the feasibility of the proposed architecture is not constrained by financial logic but by the execution model of the underlying ledger.

g) Experimental Cost Analysis and Scalability: To validate the economic feasibility of the proposed dual-layer architecture, we model the settlement cost C_{total} as a function of the total token supply S and the batch size B . In a parallel-execution environment (e.g., Solana), the cost is amortized across concurrent transactions, defined by:

$$C_{total} = \left(\frac{S}{B}\right) \times P_{tx} + \epsilon \quad (1)$$

where $P_{tx} \approx \$0.0001$ represents the base transaction fee and ϵ accounts for minor computational overhead. For a large-scale facility with $S = 10^6$ atomic units (1 Watt tokens) and a batch size $B = 10^4$, the system requires 100 concurrent transactions.

As demonstrated in Table II, this configuration results in a total settlement cost of $\approx \$0.01$ per period. This represents a 5×10^6 improvement over traditional sequential EVM models, which would incur costs exceeding $\$50,000$ for the same operation. This experimental projection confirms that asset-centric micro-distributions are not only technically viable but economically superior for high-frequency infrastructure revenue clearing.

VII. ECONOMIC AND INSTITUTIONAL RATIONALE

Traditional infrastructure finance is built on layered intermediation. Capital is aggregated through financial institutions, projects are structured through legal entities, and revenues are distributed through multiple contractual layers. Although each layer serves a function, it also introduces additional costs, complexity, and risk bundling.

In many infrastructure settings, however, the core elements of the revenue process are already defined and operational. Generation, measurement, validation, and settlement are handled by regulated or policy-backed systems. Once these processes are in place, the key question is how to connect investors directly to the resulting cash-flow stream.

The role of blockchain in this context is not to improve existing securities or replace institutional settlement systems. Its role is to provide a programmable execution layer that maps already-validated cash flows to investor-level entitlements. This enables a different structure of participation, where exposure is defined at the asset level rather than through corporate or financial intermediaries.

Instead of evaluating the full balance sheet and governance structure of an issuing entity, the investor can focus on the asset, the governing framework, and the expected cash-flow profile.

The proposed framework does not aim to replace existing financial markets. Instead, it defines a complementary structure for cases where revenues are already institutionalized and rule-based. In such settings, blockchain enables a more direct link between infrastructure assets and distributed participation, without replicating the full stack of traditional intermediation.

VIII. DISCUSSION AND SCOPE

The framework is scoped to infrastructure assets with well-defined, policy-backed cash flows. It is not a general-purpose financial system. Its effectiveness depends on reliable external settlement and validation. The contribution is representational: a direct mapping from asset-level revenues to investor participation, reducing reliance on corporate aggregation. We do not aim to replace existing financial markets but to complement them for rule-based revenue streams.

IX. CONCLUSION

This paper shows that the main challenge in RWA tokenization is not migration, but representation. Existing approaches move assets onto blockchain without adapting how they are modeled, leading to a mismatch between financial structures and execution logic.

We propose a dual-layer architecture and an asset-centric model that separates legal anchoring from economic participation. This allows asset-level cash flows to be mapped directly to programmable investor rights.

The contribution is to define a blockchain-native abstraction for representing real-world assets, focusing on cases where revenue generation and settlement are already institutionally established.

While the current dual-layer architecture successfully demonstrates asset-centric yield routing and scalable settlement, certain extensions could further enhance its expressiveness. Future work may extend this architecture by splitting the participation layer (Y-BIT) into two distinct sub-layers: one responsible for legal encumbrances (e.g., liens, pledges) and legal asset identity using a new type of token, an object-centric representation that falls under neither fungible nor non-fungible definitions, and another dedicated purely to settlement execution using a fungible token. This suggests that the existing token taxonomy may be insufficient for infrastructure assets, pointing toward a need for a third category in the literature. These refinements are left for future investigation.

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